6.2 Robotics

What are robots?

A **robot** is a mechanical device that performs an action, usually an action that a human would perform. **Robotics** is an area that covers the design, construction and operation of robots.

A robot can be controlled by a person, or they might be automated, or a mixture of the two.

One example is a drone. This is a robot, it can be controlled by a person, and it can be programmed to be perform movements depending on its surroundings.

Another example is the machines that build components in industry, for example, manufacturing devices. The robots that put the components together can be automated so that they can build the components without human interaction.

Robots have a range of features:

- A mechanical structure or framework. This is the body of the robot.
- Electrical components such as:
  - Sensors to record its environment, for example, the position of the component it is building.
  - Microprocessors, to take the reading from the sensor and decide the action to perform.
  - actuators, to make the robot move.

Robots are also **programmable**; a program can be written for the robot to follow.

KEY WORDS

robot: a machine that replicates human actions or movement.

robotics: an area of computer science that looks at the creation and use of robots.

programmable: a computer that will run the commands stored in it.

6.3 Robots in context

Robots in medicine

Developments include the use of robot nurses that can move around a hospital to visit and interact with patients. This can save the doctors and nurses time so that they can spend more time doing other tasks.

Robots are also used to perform procedures such as operations. For example, if a surgeon is not present in the hospital, even on the other side of the world, they can control the robot performing the procedure. This means that a patient will not have to wait for their procedure, and specialists are not required in every hospital because they can share resources.

The use of robots in medicine also has drawbacks. If there are any errors in the programming, or someone gains access to the programs and changes them, then the result could cause harm. If a surgeon is controlling a robot from a different location and the network connection is lost, then the procedure will stop. This means there needs to be backup systems ready to work if there are any problems.

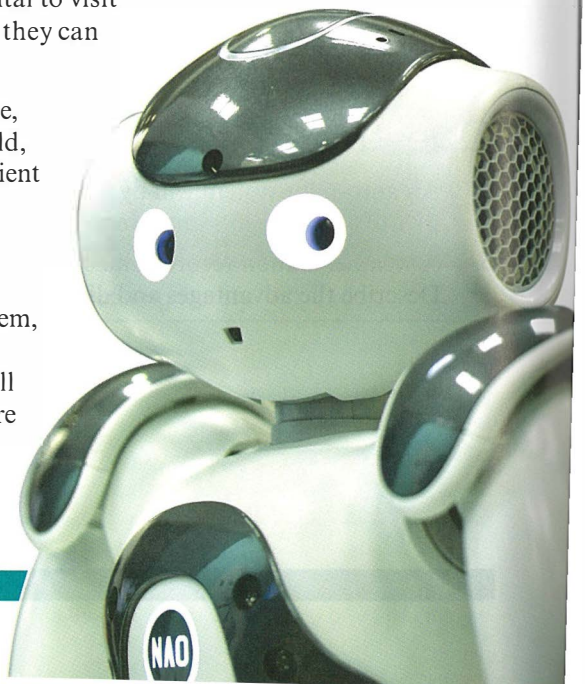




Figure 6.7: Robots preparing for surgery

Robots in agriculture

Developments include automated tractors and other machines, they can make use of satellites and satellite navigation to guide their movements and then perform the required actions. This means that a farmer can be performing other actions, whilst the machines perform the repetitive tasks that would take up their time otherwise.

Robots can also be used to plant seeds, remove weeds, check and adjust the environment where the plants are growing. All of these can be carried out 24 hours a day, 7 days a week, whereas a person may only be working 8 hours a day.

Robots in agriculture are often used for the repetitive tasks that people would otherwise spend a lot of time performing. The upfront cost might be high, and there will be maintenance required. However, it might also replace a worker who would otherwise be performing these actions.



Figure 6.8: A lawnmower working without a driver

Transport

A self-driving car is a robot. It reads its environment and moves accordingly. This is still being developed, but eventually people may not need to drive cars, this could improve safety because the robot can react faster than a person. It is an expensive up-front cost, and there is the threat of what happens if someone accesses the program of a car and changes it.

Robots are commonly used in factories to transport items. For example, in a factory, robots can move boxes of product to the lorries. These can be completely automated, keeping people away from the movement of products that could be dangerous. They do replace people's jobs, but also provide new jobs to run and maintain the system.



Figure 6.9: A self-driving car

Industry

Industry is the manufacture of goods from raw materials. This could be from the building of a pencil, to the building of an aeroplane and everything in-between.

Some factories are completely automated with robots. A car production line can involve many different robots, each performing a specific role. The car being built moves from one robot to the next without the need for human interaction. This means that the factory can run all day, every day. The robots do not need to rest and can be more precise because they are not prone to human error. However, their roles were previously carried out by people who may now be without a job. Instead a smaller number of people are required to monitor and maintain the robots, with a different skill set than those who have been replaced. The initial cost is also very large as the equipment needs to be purchased, and the program produced.

Robots don't always manage all parts of a process. Some robots can:

- Manage the packing of items, for example, putting packets of biscuits into boxes of 24 packets.
- Test products, for example, checking the temperature of products when they come out of an oven.
- Perform tasks that are minute in size and require exact precision, for example, the building of circuits.

Entertainment

There are many robots that can be bought as toys, such as robot dogs for people to interact with. There are also those that are educational, for example, where a child can build a robot and then program it to perform specific functions. These allow people to engage with technology, and in some cases learn about how robots work and be introduced to programming.

COMPUTER SCIENCE IN CONTEXT

Drones are another example of where robots are used in entertainment. People can use their drones to record videos of themselves performing tasks. These allow people to get a birds-eye view of an area, but a loss of contact between the drone and the remote could cause the drone to crash and potentially cause injury to people nearby.



Figure 6.10: A drone

Domestic

Domestic robots are those that perform tasks around a home. The most common one is the robot vacuum cleaner. It will make its way around a room, turning when it detects obstacles and changing direction. This can save people time as they will not need to perform the task themselves, however they still need monitoring and cannot go up and down steps. A robot lawnmower works in the same way, monitoring the area it is mowing to make sure it does not miss any spaces.

COMPUTER SCIENCE IN CONTEXT

A security robot can be used to protect a house. It could record what is happening as videos, and detect motion or sound. These can be set to patrol an area and alert someone if an abnormality (something unusual) is detected.

ACTIVITY 6.1

Find more examples of robots that are used in each of the sections; medicine, agriculture, transport, industry, entertainment and domestic. List the benefits and drawbacks of each of the robots. Share your findings with the class and see how many different robots you have all found.

REFLECTION

After completing Activity 6.1, it can be helpful to think about how you approached it and ways you could improve your research skills for future activities.

1 Discuss the sources you used to find examples of robots. How did you make sure these sources were reliable, up-to-date and accurate? What would you do differently if you had to find further examples?

2 Discuss whether finding your own examples might help you answer exam questions. Is it helpful to find your own contexts instead of relying on those given in class?

Questions

6 Define the term robot.

7 Identify **two** ways robots can be used in industry.

8 Give **one** benefit and **one** drawback of the use of domestic robots.

9 Discuss the use of driverless cars.

6.4 Artificial intelligence

Can computers think? This is a question that has been asked and sought after for many years. Alan Turing developed a test in 1950, called the Turing Test, to assess whether a computer can display intelligence. The test involves a person holding a conversation with a computer program, and another person. If they cannot work out which is the computer and which is the person they are talking to, then it is has passed the Turing Test. Some people do not think that this would demonstrate intelligence, the computer is still following a set of rules that are pre-written.

What is artificial intelligence?

Artificial intelligence (AI) is the development of programs to simulate human behaviour. There is a broad range of areas within AI:

- Image recognition; to identify objects or people in an image.
- Speech recognition; to identify words spoken and store them.
- Natural language; to receive a command or instruction that is not in a set format and perform the required response.
- Computer games; to move elements or characters independently based upon the environment.
- Diagnosis systems; for example, medical diagnosis.

You have probably used natural language systems, for example, on a smartphone or a web browser. You can enter your request in dozens of different ways. Let's say you want to find out what time the sun sets today. You could enter "sunset today time", or "what

KEY WORD

artificial intelligence (AI): a part of computer science that looks at creating machines that can think and perform tasks a person would usually perform.